In-Field Testing of Digital Microfluidic Biochips using Multi-Agent Reinforcement Learning

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Presentation Overview

Introduction to Digital Microfluidic Biochips (DMBs) I

- **What are DMBs?**
  Digital Microfluidic Biochips (DMBs) are programmable lab-on-chip platforms capable of manipulating discrete droplets of fluid on a 2D electrode array using electrical control.

- **Key Applications:**
  DMBs are widely used in biomedical diagnostics, point-of-care testing, DNA sequencing, and drug discovery.

- **Working Principle:**
  Droplet movement is achieved via electrowetting-on-dielectric (EWOD) phenomenon by applying voltages to control electrodes.

Introduction to Digital Microfluidic Biochips (DMBs) II

- **Why Testing Matters?**
  Ensuring reliability is critical, especially for safety-critical bioassays. In-field testing verifies functionality in real-time during assay execution.

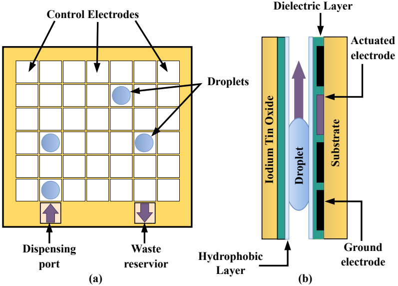

Figure 1: (a) Schematic view and (b) Side-view of a general DMB. The droplet is moving to the right by EWOD.

Related Definitions and Problem Formulation I

Related Definitions

- **Assay:** A biochemical protocol represented as a sequence of droplet operations (e.g., mixing, splitting, transporting), executed on the DMB via actuation sequences Fig. 2.
- **Snapshot:** Snapshot of the DMB at any time step t is defined as the subset of vertices on which a droplet was present.
- **Footprint Edge:** A grid edge on the DMB that is used by assay droplets during transport operations (i.e., edge that connects consecutive positions of the same droplet across adjacent snapshots).
- **Assay Footprint:** It is the union of all snapshots during assay time i.e., the set of all vertices on which a droplet was present during assay execution.

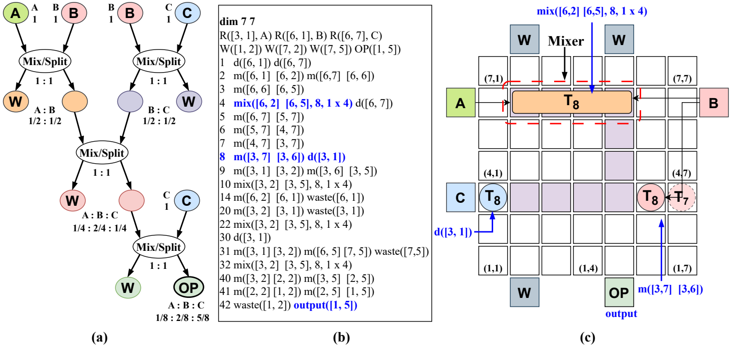

Figure 2: Mapping of a Sequential-mixing assay in 1:2:5 on a 7×7 array: (a) sequencing graph, (b) synthesized output, and (c) final mapping along with droplets position.

In-Field Testing of DMB

- **Input:**
  An actuation sequence for implementing a bioassay on the target DMB.
- **Output:**
  A test droplet routing plan, including both the path and schedule, from source to sink reservoirs.
- **Objective:**
  Determine a routing plan such that:
  - Test droplets cover assay footprint edges
  - Test droplets do not collide with assay droplets (i.e., they must avoid entering the 3×3 grid centered on any assay droplet at the same timestep).
  - The total routing time is minimized.

Literature Review I

The table below summarizes key characteristics of existing test approaches for DMBs across various criteria.

Table 1: Characteristics of Various Test Approaches for DMBs

Test Approach	**Testing Type**	**Def. Detect.**	**No. of Test Droplet**	**Scalability**	**Approach Type**
Offline Testing [Su et al., 2007][Mitra et al., 2011]	Structural	Full	1–2 (heuristic)	till 15×15	PMF/CPP
Online error detection [Mitra et al., 2012]	Functional	Partial	Nil	any size	Table-driven
In-field (concurrent) [Su et al., 2006b]	Structural (Hamiltonian)	Partial	1–2 (heuristic)	till 15×15	ILP
In-field (concurrent) [Su et al., 2007]	Structural (Eulerian)	Full	1–2 (heuristic)	till 15×15	PMF (Heuristic)
In-field (concurrent) [Bhattacharjee et al., 2017]	Structural (Eulerian)	Full	up to 4	till 15×15	SAT
Offline Testing [Dinh et al., 2015]	Structural (Eulerian)	Full	any number	till 30×30	ILP/Heuristic
In-field (concurrent) Proposed	**Structural (Eulerian)**	**Full**	**up to 4**	**till 50×50**	**Reinf. Learn.**

Literature Review II

Recently, reinforcement learning-based approaches have gained attention for scalable and adaptive droplet routing under faulty conditions.

- Liang et al. [Liang and Zhong, 2020, Liang et al., 2024] proposed an RL based approach for adaptive droplet routing based on electrode health conditions.
- Kawakami et al. [Kawakami et al., 2023] introduced a deep RL method that can dynamically adapt to both known and unknown faults during droplet routing.
- A multi-agent deep RL algorithm was developed for routing multiple droplets on micro-electrode dot array (MEDA) biochips [Zhong et al., 2018], utilizing on-chip sensing capabilities [Liang, 2021, Elfar et al., 2022].

Background of Reinforcement Learning I

- **Reinforcement Learning (RL)** is a learning paradigm where an agent learns to make decisions by interacting with an environment.
- The goal is to learn a **policy** that maximizes the cumulative reward over time.
- At each time step t, the agent:
  - observes the current state s_t,
  - selects an action a_t,
  - receives a reward r_t,
  - transitions to the next state s_{t+1}.
- The agent learns from trial and error using the reward signals to improve its actions.

Background of Reinforcement Learning II

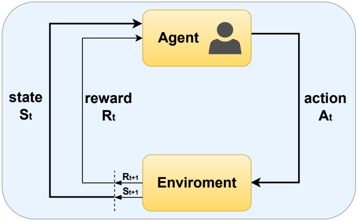

Figure 3: Agent–Environment interaction in a Markov decision process.

- **Q-Learning** is a value-based, off-policy RL algorithm.
- It learns the optimal action-value function $Q^*(s, a)$ which estimates the expected return for taking action a in state s and following the optimal policy thereafter.

Background of Reinforcement Learning III

- **Update Rule:**

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right]$$

  - α = learning rate
  - γ = discount factor
- Converges to optimal Q-values with sufficient exploration and learning iterations.
- Action selection is often done using ϵ-greedy policy.

Reinforcement Learning-Based In-Field Testing I

- **RL Algorithm Used:** Tabular Q-Learning
- **Exploration Strategy:**
  - ϵ-greedy policy
  - *Optimistic initialization* to promote early exploration
- **Single-Agent Setup:**
  - Learns optimal test droplet paths under assay constraints
  - Achieved reduced test completion time
- **Multi-Agent Extension:**
  - Extended to 4 independent agents
  - Used *Independent Q-Learning (IQL)* framework
  - **Coordination via Priority System:**
    - Priority assigned based on uncovered footprints in quadrant
    - Higher priority agent moves first
    - Others adjust actions to avoid collisions
- **Outcome:** Enhanced scalability and lower test application time while preserving assay integrity

Reinforcement Learning-Based In-Field Testing II

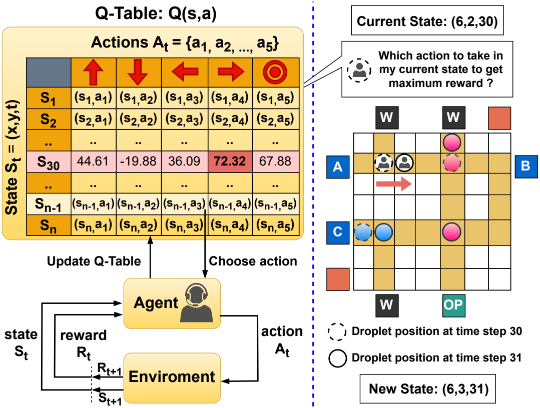

Figure 4: Single agent Q-Learning architecture. The agent updates the Q-table during training (left) and uses the optimized Q-table to navigate the DMB during deployment (right).

Reinforcement Learning-Based In-Field Testing III

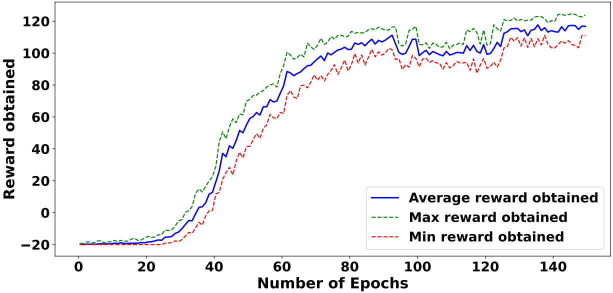

Figure 5: Training performance of the agent on Sequential-mixing assay. The graph shows the reward agent obtained per epoche.

Reinforcement Learning-Based In-Field Testing IV

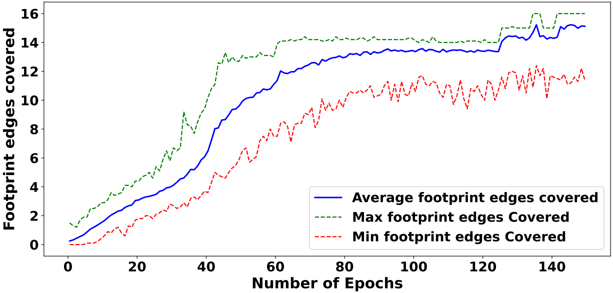

Figure 6: Footprint edges covered per epoch by the agent during training in the Sequential-mixing assay.

Reinforcement Learning-Based In-Field Testing V

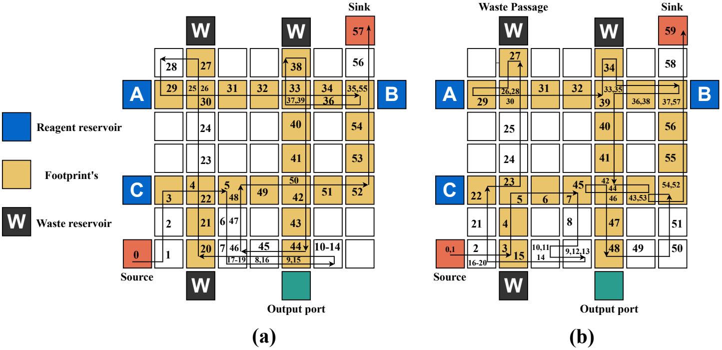

Figure 7: Test paths obtained for the Sequential-mixing assay: (a) SAT-based method, (b) Proposed method.

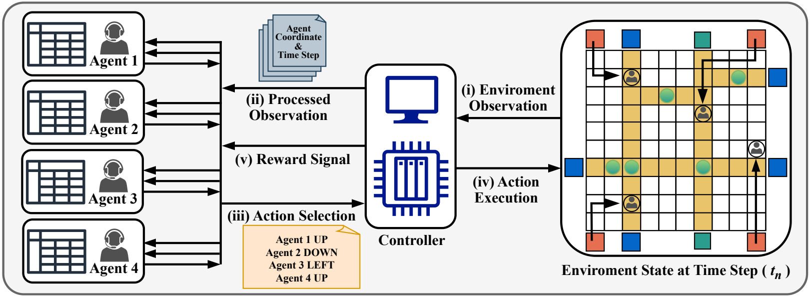

Figure 8: High level overview of the multi-agent reinforcement learning (MARL) framework for in-field testing of digital microfluidic biochips.

Experimental Setup and Results I

Table 2: Summary of Assays used for comparing different approaches

Test ID	Assay Name	Dimension	T_{assay}
Assay 1	InVitro [Bhattacharjee et al., 2017]	7 x 7	36
Assay 2	Sequential mixing of three reagents (Fig.2)	7 x 7	42
Assay 3	Multiple target mixture preparation: $A : B : C = 1 : 2 : 1$ and $A : C : D = 1 : 2 : 1$.	9 x 9	122
Assay 4	MinMix mixing [Bhattacharjee et al., 2018] $R_1 : R_2 : R_3 : R_4 = 95 : 87 : 59 : 15$	11 x 11	192
Assay 5	Multiple target mixture preparation: $A : B : C : D : E : F = 1 : 1 : 1 : 1 : 2 : 10$, and $1 : 1 : 1 : 1 : 10 : 2$	13 x 13	155
Assay 6	Multiplexed InVitro [Su et al., 2006a]	16 x 16	160
Assay 7	Three instances of PCR mixture [Roy et al., 2014]	16 x 16	220
Assay 8	Plasmid DNA [Bhattacharjee et al., 2018] mixing ratios = $(57 : 28 : 6 : 6 : 6 : 3 : 150)$	16 x 16	195
Assay 9	Multiplexed InVitro & Glucose [Su et al., 2006a, Su et al., 2007]	18 x 33	203
Assay 10	Mixing of six reagents: $A : B : C : E : G : H = 1 : 1 : 2 : 2 : 1 : 1$ and $A : B : D : F : G : H = 1 : 1 : 2 : 2 : 1 : 1$	30 x 30	184
Assay 11	Four instances of Multiplexed Glucose assays [Su et al., 2007]	32 x 32	128
Assay 12	Combined execution of five complex assays—Assay 2, Assay 4, Assay 6, Assay 8, and Assay 10	50 x 50	533

Experimental Setup and Results II

Table 3: Comparison of Experiments on Test Assays Across Methods

ID	T_{assay}	# Footprints	# Agents	Test Application Time			Running Time (seconds)			Edge Coverage %		
				SAT	ε-greedy	Opt. greedy	SAT	ε-greedy	Opt. greedy	SAT	ε-greedy	Opt. greedy
Assay 1	36	20	1	45	46	46	0.74	88.77	68.24	75.00	70.00	70.00
Assay 2	42	24	1	57	65	59	7.25	78.00	69.19	66.67	62.50	66.67
Assay 3	122	104	1	159	186	179	4831.05	1094.54	983.49	73.91	69.23	68.26
Assay 4	192	49	1	192	207	202	852.01	762.16	715.41	100	95.91	97.95
Assay 5	155	108	1	191	220	210	12698.28	1275.11	1201.97	88.04	83.33	85.18
Assay 6	160	319	4	160	207	182	19825.27	1842.81	1695.78	100.00	87.77	95.61
Assay 7	220	126	1	NA	256	252	Time Out	1396.39	1249.65	NA	92.06	92.85
Assay 8	195	115	1	NA	243	220	Time Out	1366.26	1197.09	NA	83.47	87.82
Assay 9	203	382	4	NA	277	239	Time Out	2058.34	1975.68	NA	84.03	97.12
Assay 10	184	261	4	NA	218	211	Time Out	2178.30	2094.44	NA	95.40	98.08
Assay 11	128	252	4	NA	171	151	Time Out	1673.85	1587.95	NA	84.92	90.47
Assay 12	533	768	4	NA	636	598	Time Out	8312.18	7893.27	NA	91.53	95.18

Conclusion

- The proposed RL based in-field testing framework achieves near-optimal test plans with significantly reduced computation time.
- For example, in Assay 5, RL achieves $\sim$3% less coverage than SAT but with a $\sim$10$\times$ speedup (1275s vs. 12,698s) on a 13$\times$13 grid.
- In more complex assays (Assay 7–12), SAT solvers failed to generate test plans, while RL based methods successfully produced high-coverage solutions within reasonable runtime.
- Among the two RL strategies, **optimistic Q-learning** converged faster and achieved better test application time (TAT) compared to ϵ-greedy.
- These results demonstrate that the RL based approach is scalable, efficient, and effective for in-field test generation on DMBs.

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